9) Missionaries and Cannibals using BFS :

```
from collections import deque

# State: (missionaries_left, cannibals_left, boat_side)
start = (3, 3, 0)
goal = (0, 0, 1)

moves = [(1, 0), (2, 0), (0, 1),
        (0, 2), (1, 1)]

def safe(m, c):
    mr = 3 - m
    cr = 3 - c

    return (
        0 <= m <= 3 and 0 <= c <= 3
        and (m == 0 or m >= c)
        and (mr == 0 or mr >= cr)
    )

queue = deque([(start, [])])
visited = {start}

while queue:
    state, path = queue.popleft()

    if state == goal:
        print("Solution:")
        for step in path + [state]:
            print(step)
        print("Minimum crossings:", len(path))
        break

    m, c, boat = state

    for dm, dc in moves:
        if boat == 0:
            new_state = (m - dm, c - dc, 1)
        else:
            new_state = (m + dm, c + dc, 0)

        nm, nc, _ = new_state
```

```
if safe(nm, nc) and new_state not in visited:
    visited.add(new_state)
    queue.append((new_state, path + [state]))
```